

Trades-Runtime

Multi-Fabric, Multi-Pipeline SOTA Architecture - Addendum v0.9

Scope. This addendum contains the complete architectural expansion since Workforce Capacity & Demand-Aware Scheduling v0.8. It replaces a single primary decision pipeline with a federated multi-fabric operating architecture intended to become more efficient, adaptive, fault-tolerant and operationally capable than existing trade-service platforms.

1. Architectural Thesis

Trades-Runtime should not rely on one model, one optimizer or one operational pipeline. The target architecture runs multiple independent analytical and decision pipelines against shared canonical state, reconciles their outputs, exposes disagreement and uncertainty, then selects or presents the best admissible operating action.

Canonical Event State -> N independent decision/analytic pipelines -> parallel solvers/models -> cross-model coherence -> Pareto evaluation -> policy/human constraints -> ranked decision frontier -> execution -> receipt -> settlement -> champion/challenger learning -> repeat.

2. Parallel Decision Pipelines

Pipeline	Primary role
Deterministic Rules Pipeline	Hard constraints, safety, qualification, SLA, policy and non-negotiable business rules.
Optimization Pipeline	Dispatch, routing, capacity allocation, scheduling, contribution and schedule-survival optimization.
Predictive ML Pipeline	Demand, duration, callback, inventory, conversion, urgency and capacity probabilities.
Simulation / Digital-Twin Pipeline	Counterfactual schedules, scenario testing and alternative operational futures.
LLM Reasoning Pipeline	Unstructured notes, context synthesis, explanation, root-cause hypotheses and solution generation.
Historical Analogue Pipeline	Find comparable past calls/days/branches and use empirical outcomes as decision evidence.
Cross-Branch / Federated Pipeline	Company-wide balancing of Vans, labor, inventory, booking capacity and regional support.
Human-Heuristic Pipeline	Learn recurring dispatcher/manager override patterns without assuming the human is always correct.

3. Decision Fabric

The Decision Fabric reconciles all candidate outputs instead of allowing any single pipeline to control operations.

Candidate decisions -> evidence weighting -> cross-pipeline agreement/disagreement -> hard-constraint validation -> uncertainty -> expected-value/risk scoring -> ranked action set -> human or authorized autonomous execution.

Agreement across independent methods increases confidence. Strong disagreement becomes an explicit operational signal requiring deeper comparison or human review rather than being hidden by one averaged score.

4. Multiple Functional Fabrics

Fabric	Responsibility
Real-Time Operations Fabric	Seconds/minutes: booking, dispatch, call-ins, emergencies, Van state, route rebase, employee windows.
Planning Fabric	Hours/days/weeks: staffing, PTO impact, maintenance capacity, protected reserve, branch borrowing.
Economic Fabric	Labor x cost x profit, contribution, margin, opportunity cost, pricing, procurement and warranty recovery.

Fabric	Responsibility
Evidence Fabric	Receipts, provenance, media, hashes, source conflict, point-in-time truth and reconstructability.
Learning Fabric	Model training, calibration, historical replay, skill learning, demand learning and drift detection.
Simulation Fabric	Counterfactual schedules, branch scenarios and thousands of no-impact synthetic operating runs.
Analytics Fabric	Arbitrary reports, graphs, models, scales, probabilities and human-review queries.

5. Fabric Independence with Shared Identity

Fabrics may run asynchronously and on different infrastructure, but must share canonical entity IDs, event contracts, versioning, permissions and receipt lineage. A branch, Van, appointment or inventory item must mean the same thing across every fabric.

6. Competitive Solver Architecture

For high-impact decisions, Trades should allow multiple optimizers to solve the same problem with different objective functions.

Solver	Primary objective
Customer-first	Minimize response and customer delay.
Contribution-first	Maximize expected realized contribution within constraints.
Schedule-first	Maximize survival of existing commitments and minimize cascading reschedules.
Quality-first	Maximize expected first-trip completion and minimize callback/rework risk.
Balanced multi-objective	Optimize weighted customer, quality, economics, capacity and workforce-development outcomes.
Human-imitation challenger	Predict likely expert dispatcher/manager choice as an additional comparator, not a governing truth source.

7. Solver Comparison Matrix

Each candidate plan should expose comparable outcome dimensions such as delay, contribution, on-time survival, callback risk, travel, overtime and capacity impact. Trades may select a default recommendation while preserving the alternative plans and their tradeoffs for human review.

8. Pareto Frontier Decisioning

Trades should avoid collapsing every complex operating choice into a single opaque score. For material decisions it should identify the Pareto frontier: plans where improving one important objective necessarily worsens another.

A manager may therefore see a fastest-response plan, highest-contribution plan, lowest-reschedule plan and best-balanced plan. Trades recommends one, but the alternatives remain visible and auditable.

9. Continuous Champion / Challenger Architecture

Production models should compete continuously. Each prediction or recommendation later settles against the actual outcome. New challenger models remain in Shadow until they demonstrate sufficient performance and calibration against the current champion.

Champion + Challenger A + Challenger B -> same live observations -> independent predictions -> settlement -> comparative score -> promotion, retention, recalibration or retirement.

Do not delete the prior champion immediately. Preserve rollback capability and version lineage.

10. Branch-Specific Ensembles

The best model mix may differ by branch. Trades should learn the useful contribution of local, regional and company-wide models rather than forcing one universal model.

Example: Branch 3 may perform best with a 40% local, 35% regional, 25% company ensemble, while Branch 8 may require a different mixture. Ensemble weights themselves should be calibrated and versioned.

11. Multi-Horizon Optimization

Horizon	Operating question
Seconds / minutes	Which qualified Van should receive this call right now?
2-4 hours	Will the active board collapse or require protective rebasing?
Rest of day	How should remaining work, emergency reserve and employee windows be re-optimized?
Tomorrow	What booking capacity should be exposed given roster, weather and expected demand?
7 days	Should maintenance capacity, inventory or staffing protection change?
30-90 days	Where are staffing, training, inventory or branch-support constraints forming?
6-12 months	Should the company add Vans, staffing, warehouse capacity, new branches or structural process changes?

12. Horizon Interaction Rule

Long-range optimization may inform current policy, but it cannot silently destroy present customer commitments or hard availability constraints. Immediate operational truth and constitutional rules retain precedence.

13. Multi-Fabric Fault Tolerance

The architecture should assume any model, fabric, connector or optimization service may fail independently. Real-time operations continue using surviving fabrics, deterministic rules and last-known safe state. A Simulation or Analytics outage must not stop dispatch. An AI outage must not stop booking. A local branch should continue operating if central analytics is unavailable.

14. Cross-Fabric Coherence

Cross-fabric contradictions should be surfaced explicitly. Examples: Operations says Van 214 is available while Workforce says hard-unavailable; Economic Fabric prefers a profitable call while Schedule Fabric predicts three customer displacements; Inventory says part available while Evidence Fabric marks the source stale.

TradesCoherence should compare these conflicts before the Decision Fabric produces an actionable recommendation.

15. Independent Alternate Calculation

For consequential decisions, at least one alternative method should validate or challenge the primary calculation where feasible. The alternate path should be meaningfully independent - for example, optimization vs historical analogue, ML duration forecast vs empirical cohort distribution, or model recommendation vs deterministic constraint engine.

16. Historical Replay as a Core Fabric

Every major model, policy or optimizer version should be replayed across real historical operating days before promotion. Replay sets should include ordinary days, peak-demand days, call-in days, extreme-weather days, warehouse failures, high-emergency periods and branch-borrowing events.

Compare customer delay, reschedules, first-trip completion, callback, overtime, travel, contribution, protected-capacity use and coordination failures before advancing a challenger.

17. Controlled Live Experimentation

Where operationally safe, Trades can conduct controlled branch or cohort tests after Shadow validation. Results must distinguish correlation from causal evidence and preserve treatment/control definitions, sample size, policy version and outcome receipts.

18. Measurable Industry-Leading Objective

Industry-leading status should be earned through repeated controlled outcomes rather than branding. Trades should continuously measure whether it improves the operating metrics that matter.

Metric family	Examples
Customer	Wait time, response time, on-time arrival, reschedule rate, commitment survival.
Quality	First-trip completion, callback/rework, diagnostic accuracy, workmanship outcomes.
Workforce	Overtime, utilization quality, approved-window adherence, skill coverage, training effectiveness.
Operations	Route efficiency, emergency response, protected-capacity effectiveness, schedule survival.
Economics	Realized contribution/hour, margin quality, procurement waste, warranty recovery, opportunity cost.
Inventory	Stockouts, first-trip part availability, carrying cost, transfer effectiveness.
Coordination	Chain D failure frequency, acknowledgment time, duplicate work, cross-functional delay.
Models	Calibration, prediction error, champion/challenger win rate, drift, disagreement resolution.

19. SOTA Qualification Standard

A credible SOTA claim requires sustained, reproducible performance gains across multiple branches and operating conditions. Large improvements in one metric should not count if they are purchased by worse quality, manipulated sales behavior, employee-window violations or customer displacement elsewhere.

20. Closed-Loop Operating Intelligence

Traditional field-service software primarily records and coordinates work. Trades-Runtime is intended to become a closed-loop operating intelligence system.

Observe -> normalize -> predict -> simulate -> optimize -> reconcile competing plans -> recommend/act -> receipt -> measure -> settle -> calibrate -> learn.

21. Relationship to Existing Trades Architecture

The multi-fabric design does not replace Chains A-D, Workforce Capacity, Cross-Department Matrix, Evidence Packets, Decision Gates, Universal Analytics or the Receipt Ledger. It turns those components into cooperating fabrics and competing pipelines around the same canonical company state.

22. Autonomous Action Boundary

Multiple pipelines may improve confidence, but consensus does not eliminate governance. High-consequence action still passes policy gates, evidence checks, autonomy permissions and human authority rules. A unanimous model ensemble can still be blocked by a hard rule, stale evidence or authorized human override.

23. Learning from Disagreement

Disagreement itself becomes a stored feature. Trades should learn which kinds of disagreement predict model error, which solvers dominate in particular branches/conditions and when human overrides consistently outperform all machine candidates.

24. Final Frozen Architecture

MULTI-FABRIC TRADES-RUNTIME

Canonical company/event state

- > Real-Time Operations Fabric
- > Planning Fabric
- > Economic Fabric
- > Evidence Fabric
- > Learning Fabric
- > Simulation Fabric
- > Analytics Fabric

Each fabric may run multiple deterministic, predictive, optimization, analogue and reasoning pipelines.

Their outputs converge through cross-model coherence, hard constraints, evidence quality, Pareto evaluation and policy/human control into a ranked decision frontier.

Execution produces receipts; outcomes settle the predictions; champion/challenger models learn; branch-specific ensembles recalibrate; the cycle repeats.

25. Governing Performance Target

The objective is not literal infallibility. The objective is a system that is more efficient, more adaptive, more measurable, more fault-tolerant and more operationally intelligent than existing trade-service platforms - with performance proven by real branch outcomes.

End of Trades-Runtime Multi-Fabric, Multi-Pipeline SOTA Architecture Addendum v0.9